Here a and b are known as the lower limit and upper limit of the definite integral.

Note

$\int_a^b f(x)\,dx$ is a definite constant, whereas $\int_a^x f(t)\,dt$ which is a function of the variable x.



$$\int f(x)\,dx = F(x) + c$$

$$\Rightarrow \int_a^b f(x)\,dx = F(b) - F(a)$$

Example 2.56

Evaluate $\int_0^1 \left(x^3 + 7x^2 - 5x\right)dx$

Solution:

We have already learnt about the evaluation of the integral $\int \left(x^3 + 7x^2 - 5x\right)dx$ in the previous section.

$$\therefore \int \left(x^3 + 7x^2 - 5x\right)dx = \frac{x^4}{4} + \frac{7x^3}{3} - \frac{5x^2}{2} + c$$

Now, $\int_0^1 \left(x^3 + 7x^2 - 5x\right)dx$

$$= \left[\frac{x^4}{4} + \frac{7x^3}{3} - \frac{5x^2}{2}\right]_0^1$$

$$= \left[\frac{1}{4} + \frac{7}{3} - \frac{5}{2}\right] - \left[\frac{0}{4} + \frac{0}{3} - \frac{0}{2}\right]$$

$$= \left[\frac{1}{4} + \frac{7}{3} - \frac{5}{2}\right] = \frac{1}{12}$$

Example 2.57

Find the integration for $\frac{dy}{dx} = \frac{2x}{5x^2+1}$ with limiting values as 0 and 1

Solution:

Here, $\frac{dy}{dx} = \frac{2x}{5x^2+1}$

$$\therefore y = \int_0^1 \frac{2x}{5x^2+1}\,dx$$

$$= \frac{1}{5}\int_0^1 \frac{10x}{5x^2+1}\,dx$$

$$= \frac{1}{5}\left[\log\left(5x^2+1\right)\right]_0^1$$

$$= \frac{1}{5}\left[\log 6 - \log 1\right]$$

$$= \frac{1}{5}\log 6$$

Example 2.58

Evaluate $\int_0^1 \left(e^x - 4a^x + 2 + \sqrt[3]{x}\ \right)dx$

Solution:

$$\int_0^1 \left(e^x - 4a^x + 2 + \sqrt[3]{x}\ \right)dx = \left[e^x - 4\frac{a^x}{\log a} + 2x + 3\frac{x^{\frac{4}{3}}}{4}\right]_0^1$$

$$= e - \frac{4a}{\log a} + 2 + \frac{3}{4} - 1 + \frac{4}{\log a}$$

$$= e + \frac{4(1-a)}{\log a} + \frac{7}{4}$$

Example 2.59

Evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin x\,dx$

Solution:

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin x\,dx = \left[-\cos x\right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$

$$= -\left(\cos\frac{\pi}{3} - \cos\frac{\pi}{6}\right)$$

$$= \frac{1}{2}\left(\sqrt{3} - 1\right)$$

Example 2.60

Evaluate $\int_0^{\frac{\pi}{2}} \cos^2 x\,dx$

XII Std - Business Maths & Stat EM Chapter 2.indd 43 2/28/2020 5:39:56 PM